

NEON Remote Sensing Overview

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NEON Remote Sensing Overview

SPEC School - April 3, 2026

- Bridget Hass
- Data Scientist
- Airborne Observation Platform & Data Skills

What is the National Ecological Observatory Network (NEON)?

- NSF large facility
- Continental-scale observatory
- Designed to enable understanding and forecasting of the effects of land use, climate and other environmental change
- 81 field sites
 - 47 terrestrial
 - 34 aquatic
- 180 data products

NEON's data collection methods

- IS - Instrument System
 - Fixed sensors; e.g. meteorology, soil, surface water, eddy covariance (energy fluxes).
 - Terrestrial & Aquatic
- OS - Observation System
 - Terrestrial & Aquatic
 - Data are collected manually; e.g. taxa, biomass, chemistry, microbial sequencing, pathogens
- AOP - Remote Sensing
 - Airborne Platform
 - Mobile, airborne system; hyperspectral lidar and camera data products for regional scale mapping.

NEON provides data at scale

- 81 field sites
 - 47 terrestrial
 - 34 aquatic
- Over 180 data products

Airborne Observation Platform (AOP)

- Collects airborne remote sensing data

- Covers ‘regional scale’ landscapes (min of 100 km²)
- Data products generated at high spatial resolution (≤ 1 m²)
- Waveform Lidar, Imaging Spectrometer and RGB camera

AOP Sensors

- NEON Imaging Spectrometer (NIS)
- AOP by the Numbers
 - 3 sensors
 - 30 data products (L1-L3)
 - 12 years of data (2013-2024)
 - 2-10 visits per site (7 average)
 - 200 research publications
 - 376 TB of data

AOP Flight Plan Design

AOP Sampling Collection Requirements

- Clear Skies (<10% cloud cover)
 - high quality, unobscured reflectance data
- Nominal AOP flying altitude = 1000 m AGL
 - collect data at the scale of individual plants
- Minimum 10 km x 10 km box for terrestrial sites
 - collect regional scale area around NEON sites
- Fly at peak ‘greenness’ (phenology)
 - consistency between annual collections
- Fly N-S lines, solar angles above 40°
 - consistency between flight lines*

AOP Sensors - LiDAR

LiDAR - Light Detection and Ranging

Mountain Lake Biological Station (MLBS)

Little Rock Lake, Wisconsin

Lidar Raster Data (Level 3)

Elevation Models, Canopy Height, Slope/Aspect

- DSM
- DTM
- CHM

AOP Sensors - Hyperspectral

Hyperspectral Resolution

Airborne Observation Platform Data Pipeline

NEON AOP Data Products

- Level 1
 - LiDAR Slant Range Waveform
 - Discrete Return LiDAR Point Cloud (Unclassified & Classified)

- Spectrometer Orthorectified at-Sensor Radiance
- Spectrometer Orthorectified Surface Directional Reflectance
- Spectrometer Orthorectified Surface Bidirectional Reflectance
- High-resolution Orthorectified camera imagery
- Raw data to physical units (flightline & tiled)
- Level 2
 - Vegetation Indices (ARVI, EVI, NDVI, PRI, SAVI)
 - Canopy Water Indices (MSI, NDII, NDWI, NMDI, WBI)
 - Albedo
 - Fraction of Photosynthetically Active Radiation (fPAR)
 - Leaf Area Index (LAI)
 - Canopy Nitrogen
 - Canopy Lignin
 - Total Biomass Map
 - Derived NIS products - spectral indices (flightline)
- Level 3
 - Spectrometer Orthorectified Surface Bidirectional Reflectance
 - Vegetation Indices
 - Water Indices
 - Albedo, fPAR, LAI
 - Canopy Nitrogen, Lignin, Biomass
 - Ecosystem Structure (CHM)
 - Elevation – LIDAR (DTM, DSM)
 - Slope/Aspect – LIDAR
 - High-resolution Orthorectified camera imagery mosaic
 - Derived product mosaics (1km x 1km tiles)
- ATBDs (Algorithm Theoretical Basis Documents) describe the process of data product creation

Field Spectra Data

- Field Spectra Data (DP1.30012.001)
- Leaf Clip Attachment
- Handheld Spectrometer

Working with the field spectra data product

- Explore Field Spectra Data – Scaling Ground and Airborne Observations

Recent & Future Updates to AOP data products

- 2024: BRDF and topographic-corrected reflectance product derived for the NEON spectrometer data
 - 2022-2025 AOP data now available, back-processing of earlier data has started
 - Using BRDF- and topographic-corrected reflectance to generate higher-level spectrometer data products (L2, L3) products (e.g. veg & water indices, LAI, FPAR, Albedo)
- 2025: New Lidar Sensor (Riegl VQ780-IIS)
- 2026: Mapped plant foliar trait products for NEON sites:
 - Canopy percent Nitrogen - subset available on the Portal and GEE!
 - Canopy lignin (coming soon)
- Later: Biomass

Why apply the BRDF correction?

- Corrects for ‘surface reflectance anisotropy’
- Objects look differently when viewed from different angles, and when illuminated from different locations.

UNDE 2019, Directional

UNDE 2022, Bidirectional

Foliar Trait Products - Modeling Workflow

- NEON Terrestrial Observation Sampling (TOS) Datasets
 - Geolocated foliar sampling performed for
 - * Woody individuals
 - * Herbaceous cover
 - Foliar traits
 - * Biophysical (LMA, canopy water content)
 - * Biochemical (C, N, lignin, chlorophyll, elements, etc.)
 - * Stable isotopes (δ ^{13}C , δ ^{15}N)
 - Veg structure
 - * Years $\geq$ 2020
 - Hand-delineated crown polygons
 - * Years $<$ 2020
 - Stem density + stem azimuth + crown diameter -> create crown polygons
- NEON Airborne Observation Platform (AOP) Datasets
 - Hyperspectral
 - * Level-1 reflectance data, 426 bands (385-2500 nm) at 1m resolution
 - BRDF correction
 - Topographic correction
 - Wavelength resampling
 - * Corrected reflectance
 - * Select flight lines which are least cloud-affected
 - Extract AOP Data for foliar sampling locations
 - Woody individuals
 - Herbaceous clip strips
 - Select only those 1m pixels within a tree crown which are,
 - * 1) Vegetated (NDVI $>$ threshold)
 - * 2) Not understory vegetation (CHM $>$ threshold)
 - * 3) Not affected by cloud shadows (NIR $>$ threshold)
 - For the remaining 1m pixels within a crown, perform brightness normalization on the spectra to reduce the effect of internal canopy shadows
 - Calculate average value for the spectra and lidar-derived variables for the tree crown/clip strip
 - Model inputs: 331 hyperspectral bands
 - Partial least squares regression (PLSR) modeling
 - Map foliar traits and characterize uncertainty

Canopy Nitrogen - newly available!

AOP Data Quality Considerations

- NIS - Weather Quality / Cloud Conditions

HARV weather conditions & coverage 2014-2018

HARV weather conditions 2019-2024

AOP LiDAR Sensors

	Optech Gemini (P1 & P2)	Riegl Q780 (P3)	Optech Galaxy Prime (P1 2021+)	Riegl VQ-780 II-S (P2 2025+)
Years Flow	2012-2020	2018+	2021+	2025+

	Optech Gemini (P1 & P2)	Riegl Q780 (P3)	Optech Galaxy Prime (P1 2021+)	Riegl VQ-780 II-S (P2 2025+)
Laser Pulse Length (ns)	10	3	3	3
Beam Divergence (mRad)	0.8	0.25	0.25	0.25
Nominal Pulse Repetition Frequency (PRF, kHz)	33-100	300-400 (up to 266)	150-900	270-2000 (180-1333)
Pulse Density (pulses/m ²)	2-4	4-10+	~up to 30+	~up to 50+

*The Riegl sensors use a rotating mirror so at 60 degrees only collect 2/3 of the maximum PRF

Lidar Point Density HARV 2014 v 2024 - All Points

Lidar Point Density HARV 2014 v 2024 - Ground Points

LiDAR Uncertainty

Resources for Getting Started with AOP data

AOP (& NEON) Data Access

- Data Portal - manual data download through the NEON website
 - data.neonscience.org
 - includes important documents (Quick Start Guides, Algorithm Documentation - ATBDs) and information about NEON data products
- API - programmatic download (<https://data.neonscience.org/data-api/>)
 - R (neonUtilities)
 - Python neonutilities (Python version newly available as of Fall 2024)
- Google Earth Engine (GEE)
 - Subset of data available now, continuing to add more in 2025
 - Publicly searchable/findable in the GEE Publisher Data Catalog!

AOP data on Google Earth Engine

GEE Visualization App

NEON/AOP Tutorials - GEE, Python, R

- Intro to AOP Data in Google Earth Engine (GEE) Tutorial Series
- Download and Explore NEON data
- Getting started: Download data from the Portal

Additional Educational Resources

- Workshops & Courses
- Code Hub
- Learning Hub
- <https://www.neonscience.org/resources/learning-hub>

Join us for the NEON Science Seminar Series!

- May 12, 2026: Ranjan Muthukrishnan, St. Olaf College. “Linking Local Scale Community

Upcoming Required Logins and Data Licensing Updates

- Starting in June 2026...
- Required sign-in and tokens for downloads:
 - You will need to:
 - * Create a NEON user account and sign in to download data from the Data Portal OR
 - * Use an API token to download data from the API or neonUtilities
 - Without authentication, you can still browse information about NEON resources and view limited “snapshot” visualizations, but downloads

Research Applications & Highlights

NEON data & samples inform responses to extreme events

- Lidar point clouds colorized by camera imagery at the Soaproot Saddle, CA (SOAP) site before and after the 2020 Creek Fire

NEON data & samples inform responses to extreme events

- Natural disaster events captured at 6 NEON sites + 6 other sites through the Research Support Services* program from 2012-2023
- Hurricane Michael Altered the Structure and Function of Longleaf Pine Woodlands *Kenney et al. 2021 JGR Biosciences* <https://doi.org/10.1029/2021JG006452>
- 2 more wildfires in aquatic sites since 2023: at the MCRA site in Oregon and SYCA in Arizona

Wildfire vulnerability

- Classification and clustering analysis of standing dead trees and associated park asset wildfire vulnerability in Yellowstone National Park
- Yellowstone’s Standing Dead Trees Are a Wildfire Disaster Waiting to Happen, Scientists Warn

Tree mortality

- Aboveground Biomass and Tree Mortality Revealed Through Multi-Scale LiDAR Analysis

Tree species classification: ML applications

- Individual canopy tree species maps for the National Ecological Observatory Network

Artificial Intelligence for Global Scale Mapping

- Using Artificial Intelligence to Map the Earth’s Forests

Biodiversity

- Plant diversity across dimensions: Coupling biodiversity measures from the ground and the sky

Contact

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